Caesium Vault: Software Setup & Deployment Guide

This guide provides technical specifications and step-by-step instructions for deploying the **Caesium Vault** software stack on **Zorin OS**. The deployment creates a localized edge server running offline knowledge bases, local language models, and unified reverse proxy routing.

1. Environment Preparation & Prerequisites

OS System Updates

Ensure the Zorin OS host machine is fully updated before installing stack dependencies.
sudo apt update && sudo apt upgrade -y
sudo apt install -y curl wget git build-essential ca-certificates gnupg lsb-release

Static IP Network Configuration

Assign a static IP address to the primary network interface via NetworkManager (nmcli) to prevent routing breakages on local networks.

1. Identify your active interface name:
   ip addr show

2. Apply static IP configuration (replace parameters below with your subnet details):
   sudo nmcli connection modify "Wired connection 1" \
   ipv4.method manual \
   ipv4.addresses 192.168.1.150/24 \
   ipv4.gateway 192.168.1.1 \
   ipv4.dns "1.1.1.1 8.8.8.8"
   sudo nmcli connection up "Wired connection 1"

3. Verify connectivity and address assignment:
   ip route show

2. Docker & Docker Compose Installation

Install Official Docker Engine

Set up the official Docker repository to retrieve current stable builds and the modern docker-compose-plugin.
Add Docker's official GPG key
sudo install -m 0755 -d /etc/apt/keyrings
curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor -o

```
/etc/apt/keyrings/docker.gpg
sudo chmod a+r /etc/apt/keyrings/docker.gpg

# Set up repository (Zorin OS is based on Ubuntu)
echo \
  "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.gpg]
https://download.docker.com/linux/ubuntu \
  $(. /etc/os-release && echo "$UBUNTU_CODENAME") stable" | \
  sudo tee /etc/apt/sources.list.d/docker.list > /dev/null

# Install Docker Engine and Compose Plugin
sudo apt update
sudo apt install -y docker-ce docker-ce-cli containerd.io docker-buildx-plugin
docker-compose-plugin
```

Non-Root User Privilege Management

Configure system permissions to run Docker commands without requiring sudo.

```
sudo groupadd docker
sudo usermod -aG docker $USER
```

Note: Log out and back in, or run newgrp docker to re-initialize your session permissions.

Verify current installation status:

```
docker version
docker compose version
```

3. Kiwix Server Deployment (Offline Knowledge Base)

Directory Structure Preparation

Create dedicated operational directories to store ZIM archives and service mounts under /opt/caesium.

```
sudo mkdir -p /opt/caesium/kiwix/zim
sudo mkdir -p /opt/caesium/ollama
sudo mkdir -p /opt/caesium/nginx
sudo mkdir -p /opt/caesium/webui
sudo chown -R $USER:$USER /opt/caesium
```

Fetching Offline ZIM Archives

Download required offline archives (Wikipedia, textbooks) directly into the zim repository.

```
cd /opt/caesium/kiwix/zim
# Download a compact Wikipedia ZIM dataset for testing validation
wget https://download.kiwix.org/zim/wikipedia/wikipedia_en_ray_charles_2023-06.zim -O
```

test_archive.zim

4. Local AI Model Deployment (Qwen via Ollama)

The Caesium Vault utilizes Ollama to deliver localized, zero-latency inference using lightweight quantized models.

Model Allocation Matrix

Model	Parameter Count	RAM Requirement	Storage	Use Case
Qwen 2.5	0.5B	~1.5 GB	390 MB	Low-power CPU nodes & quick queries
Qwen 2.5	1.5B	~3.0 GB	980 MB	Standard localized logic & summarization
Qwen 2.5	7.0B	~8.0 GB	4.7 GB	High-accuracy local reasoning engines

5. Reverse Proxy & Unified Routing

Unified Stack Configuration (docker-compose.yml)

Create the primary deployment file under /opt/caesium/docker-compose.yml:

```
version: '3.8'

networks:
  caesium-net:
    driver: bridge

services:
  # 1. Offline Knowledge Base (Kiwix)
  kiwix:
    image: ghcr.io/kiwix/kiwix-serve:latest
    container_name: caesium-kiwix
    restart: unless-stopped
    volumes:
      - /opt/caesium/kiwix/zim:/data
    command: "*.zim"
    networks:
      - caesium-net

  # 2. Local Language Model Engine (Ollama / Qwen)
  ollama:
    image: ollama/ollama:latest
```

```
    container_name: caesium-ollama
    restart: unless-stopped
    volumes:
      - /opt/caesium/ollama:/root/.ollama
    networks:
      - caesium-net

  # 3. Web UI Frontend for Ollama Engine
  open-webui:
    image: ghcr.io/open-webui/open-webui:main
    container_name: caesium-webui
    restart: unless-stopped
    environment:
      - OLLAMA_BASE_URL=http://ollama:11434
    volumes:
      - /opt/caesium/webui:/app/backend/data
    networks:
      - caesium-net

  # 4. Gateway Reverse Proxy (Nginx)
  nginx:
    image: nginx:alpine
    container_name: caesium-proxy
    restart: unless-stopped
    ports:
      - "80:80"
    volumes:
      - /opt/caesium/nginx/nginx.conf:/etc/nginx/nginx.conf:ro
    depends_on:
      - kiwix
      - ollama
      - open-webui
    networks:
      - caesium-net
```

Nginx Routing Engine Configuration

Create /opt/caesium/nginx/nginx.conf to direct incoming traffic from Port 80 to internal microservices based on URI paths:

```
events {
    worker_connections 1024;
}

http {
    include       mime.types;
    default_type  application/octet-stream;
    sendfile        on;
```

```
    keepalive_timeout 65;

    server {
        listen 80;
        server_name localhost;

        # Direct root to Open-WebUI interface
        location / {
            proxy_pass http://open-webui:8080;
            proxy_set_header Host $host;
            proxy_set_header X-Real-IP $remote_addr;
            proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
            proxy_set_header X-Forwarded-Proto $scheme;

            # WebSocket support for streaming responses
            proxy_http_version 1.1;
            proxy_set_header Upgrade $http_upgrade;
            proxy_set_header Connection "upgrade";
        }

        # Kiwix Knowledge Base Direct Proxy Path
        location /kiwix/ {
            proxy_pass http://kiwix:8080/;
            proxy_set_header Host $host;
            proxy_set_header X-Real-IP $remote_addr;
            proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        }

        # Raw Ollama API endpoint mapping
        location /api/ {
            proxy_pass http://ollama:11434/api/;
            proxy_set_header Host $host;
            proxy_set_header X-Real-IP $remote_addr;
        }
    }
}
```

6. Stack Initialization & Verification

Booting Services

Navigate to your configuration directory and launch all containers in detached mode:

```
cd /opt/caesium
docker compose up -d
```

Model Ingestion

Pull the specified Qwen model variant into your active Ollama instance:
```
# Pull lightweight 1.5B Qwen model variant
docker exec -it caesium-ollama ollama pull qwen2.5:1.5b
```

Verification Protocols

Confirm all backend containers are active and operational:
```
docker compose ps
```

Verify service availability via local HTTP requests:
```
# Verify Proxy Gateway
curl -I http://localhost

# Verify Ollama Engine Health
curl http://localhost/api/tags
```

Access the **Caesium Vault** interface directly by navigating to http://192.168.1.150 on any client device connected within your local subnet network.